

Overview of Cross-validation statistics

Subject: Cross-validation statistics

1.

Motivation Assume a model with one or more unknown parameters, and a data set to which the model can be fit (the training data set). The fitting process optimizes the model parameters to make the model fit the training data as well as possible. If an independent sample of validation data is taken from the same population as the training data, it will generally turn out that the model does not fit the validation data as well as it fits the training data. The size of this difference is likely to be large especially when the size of the training data set is small, or when the number of parameters in the model is large. Cross-validation is a way to estimate the size of this effect.

2.

k-fold cross-validation In k-fold cross-validation, the original sample is randomly partitioned into k equal sized subsamples, often referred to as "folds". Of the k subsamples, a single subsample is retained as the validation data for testing the model, and the remaining $k - 1$ subsamples are used as training data. The cross-validation process is then repeated k times, with each of the k subsamples used exactly once as the validation data. The k results can then be averaged to produce a single estimation. The advantage of this method over repeated random sub-sampling (see below) is that all observations are used for both training and validation, and each observation is used for validation exactly once. 10-fold cross-validation is commonly used, but in general k remains an unfixed parameter. For example, setting $k = 2$ results in 2-fold cross-validation. In 2-fold cross-validation, the dataset is randomly shuffled into two sets d0 and d1, so that both sets are equal size (this is usually implemented by shuffling the data array and then splitting it in two). We then train on d0 and validate on d1, followed by training on d1 and validating on d0. When $k = n$ (the number of observations), k-fold cross-validation is equivalent to leave-one-out cross-validation. In stratified k-fold cross-validation, the partitions are selected so that the mean response value is approximately equal in all the partitions. In the case of binary classification, this means that each partition contains roughly the same proportions of the two types of class labels. In repeated cross-validation the data is randomly split into k partitions several times. The performance of the model can thereby be averaged over several runs, but this is rarely desirable in practice. When many different statistical or machine learning models are being considered, greedy k-fold cross-validation can be used to quickly identify the most promising candidate models.

3.

General case In most other regression procedures (e.g. logistic regression), there is no simple formula to compute the expected out-of-sample fit. Cross-validation is, thus, a generally applicable way to predict the performance of a model on unavailable data using numerical computation in place of theoretical analysis.

4.

k*l-fold cross-validation This is a truly nested variant which contains an outer loop of k sets and an inner loop of l sets. The total data set is split into k sets. One by one, a set is selected as the (outer) test

set and the $k - 1$ other sets are combined into the corresponding outer training set. This is repeated for each of the k sets. Each outer training set is further sub-divided into l sets. One by one, a set is selected as inner test (validation) set and the $l - 1$ other sets are combined into the corresponding inner training set. This is repeated for each of the l sets. The inner training sets are used to fit model parameters, while the inner test set is used as a validation set to provide an unbiased evaluation of the model fit. Typically, this is repeated for many different hyperparameters (or even different model types) and the validation set is used to determine the best hyperparameter set (and model type) for this inner training set. After this, a new model is fit on the entire outer training set, using the best set of hyperparameters from the inner cross-validation. The performance of this model is then evaluated using the outer test set.

5.

Limitations and misuse Cross-validation only yields meaningful results if the validation set and training set are drawn from the same population and only if human biases are controlled. In many applications of predictive modeling, the structure of the system being studied evolves over time (i.e. it is "non-stationary"). Both of these can introduce systematic differences between the training and validation sets. For example, if a model for prediction of trend changes in financial quotations is trained on data for a certain five-year period, it is unrealistic to treat the subsequent five-year period as a draw from the same population. As another example, suppose a model is developed to predict an individual's risk for being diagnosed with a particular disease within the next year. If the model is trained using data from a study involving only a specific population group (e.g. young people or males), but is then applied to the general population, the cross-validation results from the training set could differ greatly from the actual predictive performance. In many applications, models also may be incorrectly specified and vary as a function of modeler biases and/or arbitrary choices. When this occurs, there may be an illusion that the system changes in external samples, whereas the reason is that the model has missed a critical predictor and/or included a confounded predictor. New evidence is that cross-validation by itself is not very predictive of external validity, whereas a form of experimental validation known as swap sampling that does control for human bias can be much more predictive of external validity. As defined by this large MAQC-II study across 30,000 models, swap sampling incorporates cross-validation in the sense that predictions are tested across independent training and validation samples. Yet, models are also developed across these independent samples and by modelers who are blinded to one another. When there is a mismatch in these models developed across these swapped training and validation samples as happens quite frequently, MAQC-II shows that this will be much more predictive of poor external predictive validity than traditional cross-validation. The reason for the success of the swapped sampling is a built-in control for human biases in model building. In addition to placing too much faith in predictions that may vary across modelers and lead to poor external validity due to these confounding modeler effects, these are some other ways that cross-validation can be misused: